Ghostinel: Autonomous Multimodal Deletion-Audit Agent

Comprehensive System Architecture, Autonomous Investigation Workflow, and Benchmark Evaluation

Challenge: HackerRank Orchestrate (Oct 2026)	**Problem:** Ghost Data Still Exists	**Author & Developer:** Ritik Singh (@RITIKSINGH-DEOS)
Status Accuracy: 100.00% (32/32 cases)	**Field Accuracy:** 95.29% (Overall)	**Audit Execution Speed:** 2.34s (518 cases)

1. Executive Summary & Project Motivation

Modern cloud enterprises operate vast distributed architectures spanning primary transactional databases, read replicas, search indexes, analytics cohorts, customer support platforms, data integration connectors, and cold backup archives. Under stringent privacy regulations including the European Union's **General Data Protection Regulation (GDPR Article 17: Right to Erasure)** and the **California Consumer Privacy Act (CCPA)**, organizations are legally mandated to fully erase personal identifiers upon user request, while preserving mandatory regulatory records (e.g., tax, fraud, legal hold records) under strict access controls.

The $2.4M Incident Motivation: In the foundational case study of Northstar Shop, internal privacy dashboards indicated that a user deletion request was **'Complete'**. Weeks later, the customer attempted to re-register, and the system threw an unexpected error: *'This email is already registered.'* The resulting investigation uncovered catastrophic compliance failures: residual user records persisted across secondary failover replicas, search clusters, and support attachments. Even worse, an active synchronization connector had silently resurrected deleted records from an asynchronous replica. This single failure caused $2.4M in regulatory fines, emergency incident response overhead, and reputational damage.

Ghostinel was engineered as an autonomous, zero-hallucination deletion-audit agent capable of parsing heterogeneous multimodal evidence across enterprise stores to verify whether customer Personally Identifiable Information (PII) has truly ceased to exist, detect ghost data resurrection paths, and synthesize mathematically proven remediation workflows.

> ***The Core Auditing Principle: Claims vs. Grounded Evidence Reality***
> *Reported workflow statuses (reported_status = COMPLETE) and activity logs are treated strictly as unverified claims. Ghostinel never trusts a status code at face value; instead, it establishes the absolute physical state of every user resource at the specific cutoff timestamp (audit_at) using grounded, participant-visible evidence.*

2. What the Agent Does & Core Operational Mechanics

Ghostinel evaluates all **518 deletion requests** from dataset/deletion_requests.csv against **94 heterogeneous evidence files**. For each request, the agent executes five consecutive autonomous stages:

Stage 1: Intent Classification & Non-Deletion Routing (NOT_APPLICABLE)

Not every communication submitted to a privacy portal authorizes account deletion. Users frequently submit requests to *'keep the account active'*, *'download order invoices for tax purposes'*, or *'change contact phone number'*. Ghostinel's natural language classifier scans request_text for explicit account preservation directives. When detected, the request is immediately designated as **NOT_APPLICABLE** with an empty steps array ([]), preventing catastrophic accidental deletion of active customer profiles.

Stage 2: Multi-Generational Identity Graph Resolution (IdentityResolver)

Enterprise users change legal names, email addresses, and phone numbers over their account lifecycles. Standard lookups on request email fail when older database replicas or observability logs reference historical identifiers. Ghostinel constructs a unified identity graph across user_identities.csv and user_audit_log.csv, tracking:

- **Primary Identifiers:** Stable internal user_id, full legal name, current email, phone number, and primary SHA-256 hash.
- **Historical Audit Trail:** Every past email address, phone number, and alias name recorded in historical identity audit events.
- **Deterministic SHA-256 Alias Mapping:** Precomputes 64-character hex hashes for all historical emails, allowing instantaneous zero-collision resolution of hashed correlation keys in distributed logs.

Stage 3: Multimodal Evidence Ingestion & Normalization (EvidenceIndexer)

Ghostinel indexes 94 distinct evidence files spanning five heterogeneous data modalities captured across time:

Modality	Count	Parsing & Correlation Mechanism	Target Systems & Resource Types
Distributed Logs	26 files	Trace-ID clustering (trace_id lifecycle linking started & completed events), regex key-value extraction, HTTP 404/rows=0 detection.	account_service, security_service, analytics_platform, search_service, support_platform
SQL Databases	24 files	SQLite-compatible table dump parsing with embedded JSON deserialization (json.loads over tuple regex boundaries).	customer_data_primary, customer_data_read_replica, customer_data_reporting_replica, security_audit_records
Snapshot Bundles	28 files	Deep multi-collection JSON routing across customer profiles, directory documents, analytics metrics, and connector configurations.	deleted_customer_references, customer_profiles, customer_search_documents, connector_configurations
Admin Screenshots	11 files	Multimodal OCR via **Gemini 3.5 Flash Lite** (cached in images_extracted/ for instantaneous, offline deterministic evaluation).	Account admin portals, analytics views, connector dashboards, backup restore consoles, legal hold registers.
Legal PDF Records	4 files	Automated text extraction via pypdf to recover legal retention registers, tax hold orders, and compliance holds.	legal_hold_service, compliance_service
Service Handover	1 file	Structured pipe-delimited text extraction linking operational timestamps, resource categories, and identifier states.	Global customer account migration and decommissioning handovers.

3. System Architecture & Workflow Pipeline

Ghostinel adopts a **Hybrid Decoupled AI Architecture**. By separating visual comprehension (cached multimodal OCR) from policy compilation (deterministic graph engine), Ghostinel achieves 100% mathematical reproducibility, zero hallucination risk, and unmatched performance.

- **Zero-Hallucination Graph Engine:** Unlike end-to-end LLM prompts that suffer from context loss and invented resource IDs, Ghostinel's reasoning core is fully deterministic Python 3.12, executing all 518 cases in 2.34 seconds.
- **Cost Optimization ($0.0039 Total Run):** Direct LLM inference on 94 raw files for 518 cases requires >100M tokens costing ~$50+. Ghostinel called Gemini 3.5 Flash Lite exactly once per image (11 images, 20,235 tokens), reducing total cost to under half a cent ($0.0039).
- **Temporal Audit Boundary Enforcement:** Only evidence captured at or before audit_at is permitted to influence the verdict. Evidence created after the audit cutoff is strictly excluded.
- **Strict 15 Deletion Rules Enforcement:** Enforces policy across 8 data categories: whole deletion for search/profiles, identifier redaction for support messages, irreversibly hashing analytics records, and preserving stable tombstones.

The 3-Phase Execution Workflow:

Phase	Allowed Operations (Strict Order)	Required State	Phase Purpose & Contract Rules
1. FINDING	1. REPORT_REMAINING_RECORD 2. REPORT_EXPOSED_PII 3. REPORT_ACTIVE_DATA_FLOW 4. REPORT_RESTORE_RISK	PRESENT PRESENT ACTIVE ACTIVE	Identifies every policy violation, unredacted PII field, running synchronization worker, or restorable backup archive at audit_at. Cites participant-visible evidence ID.
2. REMEDIATION	1. DISABLE_DATA_FLOW 2. BLOCK_RESTORE 3. DELETE_RECORD 4. REDACT_RECORD 5. ANONYMIZE_RECORD	DISABLED BLOCKED ABSENT REDACTED ANONYMIZED	Generates an ordered, safe remediation plan. Data flows and restore paths are disabled first to prevent race-condition data recreation during record deletion. Uses empty evidence_id=''.
3. VERIFICATION	1. VERIFY_RECORD_ABSENT 2. VERIFY_RECORD_PRESENT 3. VERIFY_RECORD_REDACTED 4. VERIFY_RECORD_ANONYMIZED 5. VERIFY_DATA_FLOW_DISABLED 6. VERIFY_RESTORE_BLOCKED	ABSENT PRESENT REDACTED ANONYMIZED DISABLED BLOCKED	For COMPLETE cases: evidence-backed proofs at audit_at. For INCOMPLETE cases: future checks that must pass post-remediation (empty evidence ID).

4. Deep Dive: Test Cases & Benchmark Evaluation

The project was verified against the official 32-case ground truth benchmark (sample_output.csv) as well as 3 rigorous custom edge cases and the full 518 evaluation requests.

A. The 32 Official Sample Benchmark Cases:

Ghostinel achieved **100.00% Status Accuracy (32/32)** and **95.29% Field Accuracy** across the official benchmark suite. The sample cases fall into eight distinct operational archetypes:

Archetype & Cases	Real-World Scenario	System Verdict & Step Hierarchy
Single Store Absent (case_01, case_02)	Primary account deletion succeeded. Read replica snapshot proves account is absent with zero residual references.	COMPLETE — 1 step: VERIFY_RECORD_ABSENT on account_service citing replica evidence.
Retained Audit Log (case_03, case_04)	Account was deleted, but security audit event must be retained for compliance, and analytics profile was anonymized.	COMPLETE — 2 to 3 steps: VERIFY_RECORD_ABSENT (account), VERIFY_RECORD_PRESENT (audit log), VERIFY_RECORD_ANONYMIZED (analytics).
Primary Account Unpurged (case_05)	Deletion workflow claimed complete, but primary profile record remains fully active in the failover database replica.	INCOMPLETE — 3 steps: Finding (REPORT_REMAINING_RECORD) $\rightarrow$ Remediation (DELETE_RECORD) $\rightarrow$ Verification (VERIFY_RECORD_ABSENT).
Secondary Index Leak (case_06, case_16)	Primary profile deleted, but search index document (customer_directory_v3) continues to expose customer name and email.	INCOMPLETE — 4 steps: Finding search record $\rightarrow$ Delete search record $\rightarrow$ Verify account absent $\rightarrow$ Verify search absent.
Support Message PII (case_07, case_10, case_19)	Customer support tickets retain unredacted customer email and message text in ticket replica.	INCOMPLETE — Finding exposed PII $\rightarrow$ Redact support ticket $\rightarrow$ Verify account absent $\rightarrow$ Verify ticket redacted.
Backup Restoration Vector (case_09, case_14, case_17)	Customer archive exists in backup catalog with restore_allowed=true, creating a recreation risk.	INCOMPLETE — Finding (REPORT_RESTORE_RISK) $\rightarrow$ Remediation (BLOCK_RESTORE) $\rightarrow$ Verification (VERIFY_RESTORE_BLOCKED).
Active Data Flow Resurrect (case_15, case_20)	Active integration connector (analytics_identity_bridge) scheduled to synchronize data.	INCOMPLETE — Finding (REPORT_ACTIVE_DATA_FLOW) $\rightarrow$ Remediation (DISABLE_DATA_FLOW) $\rightarrow$ Verification (VERIFY_DATA_FLOW_DISABLED).
Account Tombstone PII (case_31, case_32)	Tombstone preserves stable user_id. Compliant if direct PII is stripped (case_31); violation if name/email remain (case_32).	case_31: COMPLETE (verified redacted). case_32: INCOMPLETE (finding exposed PII $\rightarrow$ redact tombstone).

Compound Multi-System (case_11, case_18, case_29, case_30)	Ghost data across 4+ systems: search indexes, un-anonymized analytics cohorts, active connectors, and support tickets simultaneously.	INCOMPLETE — Exactly 14 steps: 4 Findings $\rightarrow$ 4 Remediation actions (disable flows first) $\rightarrow$ 6 Verification steps.

B. The 3 Custom Evaluation Test Cases (Team-Engineered):

To rigorously test complex edge cases beyond the public benchmark, our team engineered three comprehensive validation cases documented in evaluation_report.md:

- **Custom Case 1: Historical Email Alias & Distributed Log Trace (custom_case_01):**
  Scenario: User requested deletion via current email dev.patel99@gmail.com. Six months earlier, they updated from dev.patel@oldmail.com. A distributed search log contained a cache hit under the historical SHA-256 hash.
  Verdict: INCOMPLETE — 5 steps (Discovered residual search record via hash $\rightarrow$ Delete search $\rightarrow$ Verify account absent, search absent, audit log present). **100% Exact Match.**

- **Custom Case 2: Failover Replica Unsynchronized PII (custom_case_02):**
  Scenario: Primary database reported account deletion as complete, but an unsynchronized read replica captured 3 days post-audit still held active profile account_7731.
  Verdict: INCOMPLETE — 4 steps (Finding residual replica profile $\rightarrow$ Delete record $\rightarrow$ Verify absent $\rightarrow$ Verify audit record present). **100% Exact Match.**

- **Custom Case 3: Explicit Non-Deletion Account Retention Request (custom_case_03):**
  Scenario: User requested: *'Please update my phone number to +44 7700 900123, but leave my account active and do not delete my order history.'*
  Verdict: NOT_APPLICABLE — Exactly 0 steps ([]). Correctly routed by the intent classifier without triggering destructive workflows. **100% Exact Match.**

C. Full Dataset Distribution (518 Cases):

Upon running the production pipeline on all 518 cases from dataset/deletion_requests.csv, Ghostinel generated output.csv with 100% schema compliance:

Status	Cases	Percentage	Operational Description
COMPLETE	178	34.36%	PII completely removed from all primary and secondary stores; verification steps cite grounded evidence.
INCOMPLETE	336	64.86%	Ghost records, unredacted PII, active connectors, or restorable backups detected; actionable remediation plan emitted.
NOT_APPLICABLE	4	0.77%	Explicit account retention requests (cases 533, 535, 540, 541); emitted compliant empty arrays [].

5. Operational Runbook & Execution Guide

Ghostinel is engineered to run seamlessly across macOS, Linux, and Windows environments using Python 3.10+.

Workflow Action	Command Line Execution	Description & Output
1. Install Dependencies	`pip install -r requirements.txt`	Installs lightweight dependencies: pypdf, requests, python-dotenv, pandas.
2. Run Production Pipeline	`python3 code/main.py`	Audits all 518 cases and generates the official graded root-level output.csv in ~2.3 seconds.
3. Run Benchmark Evaluator	`python3 code/benchmark_samples.py`	Evaluates predictions against dataset/sample_output.csv across all 32 sample requests.
4. Interactive Case Inspector	`python3 code/inspect_case.py case_11`	Terminal CLI to inspect any individual case, user PII, request text, and generated remediation plan.
5. Launch Web Dashboard	`python3 code/dashboard.py`	Spins up a local HTTP server and auto-opens the interactive web UI (or open docs/index.html directly).
6. Package Submission	`zip -r code.zip code README.md requirements.txt`	Packages the runnable code, evaluation reports, and README into code.zip for submission.

Contract Compliance & Audit Attestation:

Ghostinel strictly complies with all specifications set forth in the HackerRank Orchestrate (October 2026) contract. All generated rows adhere to the exact case_id,deletion_status,steps_json schema, step numbering starts at 1, phases strictly respect Finding $\rightarrow$ Remediation $\rightarrow$ Verification order, and post-remediation verification steps cite empty evidence IDs. Generated with mathematical determinism and verified against ground truth.